Gukyeong Kwon

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Summary

Research scientist with 5+ years of industry experience leading multimodal foundation model research at Amazon. Led vision encoder training and data mixture optimization for the Nova family, spanning multimodal understanding and universal retrieval. Expertise in compute-efficient training, cross-modal alignment, and scaling ladder design. Area Chair for ECCV; published at ICLR, ECCV.

Work Experiences

Senior Applied Scientist at Amazon AGI, New York, NY — October 2023 - Present

Led a team of 10 scientists and engineers for Nova Multimodal Embeddings; partnered with PMs to translate customer requirements into science roadmaps and allocated research workstreams across the team. Mentored 3 research interns (ACL Findings, ECCV 2026 submission).

- **Multimodal Foundation Model: Visual Token Efficiency and Data Mixture Optimization**
  - Designed the multi-stage vision encoder training curriculum — compute-efficient contrastive learning in early stages, followed by end-to-end training with the vision encoder unfrozen and connected to the LLM, which proved essential for understanding information-dense inputs such as documents and infographics.
  - For the early contrastive stage, achieved equivalent classification and VQA performance at 25% of the compute budget through optimal token budget distribution and resolution-constrained curricula across training sub-stages.
  - Systematically evaluated token compression approaches (attention-based, learnable, statistics-based); identified statistics-based merging as dominant, reducing output tokens by 75% with minimal degradation.
  - Optimized multimodal data mixtures across pre-training and SFT stages; developed a scaling-based methodology to identify per-dataset contribution and saturation, informing token allocation for under-trained data types.

- **Universal Multimodal Embeddings: Cross-Modal Alignment and Retrieval**
  - Achieved leading retrieval accuracy across five modalities (image, text, document, video, audio) for Amazon Nova Multimodal Embeddings through systematic data mixture optimization and multi-stage training.
  - Developed ranking-aware knowledge distillation for contrastive learning using teacher similarity scores as soft supervision rather than representation fidelity, based on the insight that ranking matters more than embedding precision for retrieval; particularly effective for improving text-to-text retrieval performance.
  - Established scaling ladders to identify optimal batch size, learning rate, and model size; systematically balanced compute budget against retrieval accuracy given contrastive learning's sensitivity to global batch size.
  - Trained vision encoders with native resolution support by investigating positional embedding interpolation strategies and data packing, enabling efficient handling of variable aspect ratios without resolution-dependent artifacts.
  - Mitigated intra-modality retrieval bias through diversified instruction tuning, enforcing cross-modal alignment while minimizing per-modality performance trade-offs.

Applied Scientist at AWS AI Labs, New York, NY — January 2021 - October 2023

- **MaskVLM: Masked Vision and Language Modeling**
  - Proposed MaskVLM, a cross-modal masked reconstruction method that learns vision-language alignment by reconstructing one modality's masked signal conditioned on the other (ICLR 2023).
  - Achieved state-of-the-art on zero-shot text-to-image retrieval (+6.8 over ALBEF), VQA, visual reasoning, and visual entailment; outperformed ALBEF by 8.7 points on image retrieval under a 1M-pair training budget.
  - Provided theoretical interpretation through Variation of Information (VI), showing that bidirectional conditional modeling minimizes VI to more accurately estimate joint distributions.
- **Image - Text Embeddings: Web Data Preparation and Large-scale Contrastive Learning**
  - Curated billion-scale web training data through topic-based balancing and synthetic caption enrichment; stochastic sampling of synthetic captions during training improved mean Recall@1 on 6 retrieval benchmarks ($0.38 \rightarrow 0.46$).
  - Designed and benchmarked multi-encoder architectures (ViT/Swin $\times$ T5/RoBERTa) at scale, identifying ViT + RoBERTa as optimal given ViT's compatibility with masked pre-training, superior training stability, and broader availability of community pre-trained weights for initialization.

- Applied Matryoshka representation learning to support multiple embedding dimensions (1024, 384, 256) in a single model, enabling flexible accuracy-latency trade-offs at deployment without retraining.

Graduate Research Assistant, Georgia Tech, Atlanta, GA — August 2016 – August 2021

- **Anomaly Detection Using Gradient-Based Representations**
  - Developed gradient-based representations for anomaly detection grounded in Fisher kernel theory from information geometry; achieved state-of-the-art OOD detection on MNIST (0.97 AUROC) and fMNIST (0.93 AUROC), published at ECCV 2020 and ICIP 2019 (Best Paper Award, top 0.1%).
  - Developed a gating model for generalized zero-shot learning that calibrates bias toward seen classes, extending the autoencoder design to improve classification on both seen and unseen categories (IEEE TIP 2022).

Selected Publications

- G. Kwon, Z. Cai, A. Ravichandran, E. Bas, R. Bhotika, and S. Soatto, "Masked Vision and Language Modeling for Multi-modal Representation Learning," *International Conference on Learning Representations (ICLR),* 2023. [arXiv]
- Amazon Artificial General Intelligence, "Amazon Nova Multimodal Embeddings: Technical report and model card," *Amazon Science,* 2025. (Led training strategy, data mixture optimization across five modalities, and scaling) [Link]
- Amazon Artificial General Intelligence, "The Amazon Nova Family of Models: Technical Report and Model Card," *Amazon Science,* 2024. (Led vision encoder training) [arXiv]
- G. Kwon, M. Prabhushankar, D. Temel and G. AlRegib, "Backpropagated Gradient Representations for Anomaly Detection," In *European Conference on Computer Vision (ECCV),* 2020. [arXiv] [GitHub] [Video] [Slides]
- G. Kwon*, M. Prabhushankar*, D. Temel and G. AlRegib, "Distorted Representation Space Characterization Through Backpropagated Gradients," *2019 IEEE International Conference on Image Processing (ICIP), Taipei, Taiwan, 2019.* (*: equal contribution, ***Best Paper Award (top 0.1%)***) [arXiv] [GitHub] [Poster] [Award]
- G. Kwon and G. AlRegib, "A Gating Model for Bias Calibration in Generalized Zero-shot Learning," *IEEE Transactions on Image Processing,* 2022. [arXiv] [GitHub]
- Z. Cai, G. Kwon, A. Ravichandran, E. Bas, Z. Tu, R. Bhotika, and S. Soatto, "X-DETR: A Versatile Architecture for Instance-wise Vision-Language Tasks," *European Conference on Computer Vision (ECCV),* 2022. [arXiv]
- X. Fu, S. Zhang, G. Kwon, P. Perera, H. Zhu, Y. Zhang, A. Li, W. Wang, Z. Wang, V. Castelli, P. Ng, D. Roth, B. Xiang, "Generate then Select: Open-ended Visual Question Answering Guided by World Knowledge," *The 61st Annual Meeting of the Association for Computational Linguistics (ACL) Findings,* 2023. [arXiv]

Education

Georgia Institute of Technology — Atlanta, GA
Ph.D. in Electrical and Computer Engineering — *August 2015 – August 2021*
M.S. in Electrical and Computer Engineering — *August 2015 – May 2018*

Sungkyunkwan University (SKKU) — Suwon, South Korea
B.S. in Electronic and Electrical Engineering — *March 2009 – August 2015*

Non-provisional Patents

- A. Parchami, G. AlRegib, D. Temel, M. Prabhushankar, and G. Kwon, "Variance of gradient based active learning framework for training perception algorithms," US 12,079,738 B2. [Link]
- G. AlRegib, G. Kwon, M. Prabhushankar, and D. Temel, "Detecting and Classifying Anomalies in Artificial Intelligence Systems," US 2022/0327389 A1. [Link]

Skills

- **Distributed training**: FSDP, DeepSpeed, Megatron | **Programming**: Python, PyTorch
- **Modeling**: Multimodal transformers, contrastive learning, knowledge distillation, vision-language models
- **Infrastructure**: AWS (SageMaker, EC2, S3, EMR), large-scale data processing
- **Research areas**: Compute-efficient training, data mixture optimization, cross-modal alignment, scaling ladder design, multimodal retrieval, token efficiency

Professional Services

- Area Chair for the European Conference on Computer Vision (ECCV).
- Reviewer for CVPR, Transactions on Image Processing, Elsevier Signal processing: Image Communication, ICIP, and ICASSP (Recognized as an Outstanding reviewer in CVPR 2023 and 2025).